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GIET UNIVERSITY, GUNUPUR – 765022

B. Tech (Sixth Semester) :: Question Bank

Subject Code – Subject Name

(Branch)

UNIT - I

PART – A: (Short Answer Questions) (2Marks)

		[CO#]	[PO#]
a.	What do you understand by Machine learning?		
b.	Discuss with examples some useful applications of machine learning.		
c.	What are the three stages to build the hypotheses or model in machine learning?		
d.	What is the difference between inductive and deductive learning?		
e.	Differentiate between Hypothesis Space and Version Space.		
f.	What do you mean by hypothesis space, instance space, version space?		
g.	Discuss with examples some useful applications of machine learning.		
h.	Explain the concept of Inductive Bias.		
i.	Distinguish between parametric and non-parametric learning.		
j.			

PART – B: (Long Answer Questions) (10Marks)

		Marks	[CO#]	[PO#]
1. a.	What is the problem of Concept Learning? State with the help of an example.	5		
b.	Why concept learning task is stated as a space search problem?	5		
2.	Apply Find-S algorithm on Training Data-I to list the maximally specific esis.	10		

Training Data-I: Positive and negative training examples for the target concept *EnjoySport*

Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

3.	Apply Candidate Elimination Algorithm on Training Data-II to list the version space.	10		

Training Data-II: Positive and negative training examples for the target concept *PackagingNeeded*

Example	Size	Color	Shape	Class/Label
1	Big	Red	Circle	No
2	Small	Red	Triangle	No
3	Small	Red	Circle	Yes
4	Big	Blue	Circle	No
5	Small	Blue	Circle	Yes

4. a.	What do you mean by a well–posed learning problem? Explain the important	5		

	that are required to well–define a learning problem.			
b.	Describe in detail all the steps involved in designing a learning system.	5		
5. a.	Define Concept and Concept Learning. With example explain how the Concept g task determines the Hypothesis for given target concept.	5		
b.	Discuss Concept learning as search with respect to General to specific ordering thesis.	5		
6. a.	Explain the candidate elimination algorithm	5		
b.	Describe List then Eliminate Algorithm	5		

UNIT - II

PART – B: (Short Answer Questions) (2 Marks)

		[CO#]	[PO#]
a.	What is "Curse of Dimensionality"?		
b.	Distinguish between Feature Selection and Feature Extraction.		
c.	Define the problem Multi Collinearity.		
d.	Distinguish between Bias and Variance.		
e.	Distinguish between Overfitting and Underfitting.		
f.	Distinguish between Regression and Classification.		
g.	Why Regularization techniques are needed in Machine Learning?		
h.	What is feature scaling?		
i.	Distinguish between Filter and Wrapper methods.		
j.	List out five dimensionality reduction techniques used in ML.		
k.	How to identify if the model is overfitting or underfitting?		
l.	What is adjusted R-Squared		

PART – B: (Long Answer Questions) (10 Marks)

		Marks	[CO#]	[PO#]
1. a.	Analyse how logistic regression can be used for classification.	5		
b.	What is subset selection problem? Distinguish between forward and backward step wise selection algorithm.	5		
2. a.	For a simple linear regression, obtain the expression for sum of square of errors (SSE).	5		
b.	What is the difference between Simple Linear and Multiple Linear Regression?	5		
3. a.	What is Multi-Variate linear regression? Explain with an example?	5		
b.	Discuss about the metrics used for evaluating a Regression Model.	5		
4.	The sales of a company (in million dollars) for each year are shown in the table below:			

x (year)	2005	2006	2007	2008	2009
Y (sales)	12	19	29	37	45

a.	Find the least square regression line y = a + bx	5		
b.	Use the least square regression line as a model to estimate the sales of the company in 2012.	5		
5. a.	Explain PCA and its process with their applications.	5		
b.	How Matrix factorization works in PCA. Explain in detail?	5		
6. a.	Explain with the help of a neat block diagram how multi variate linear regression (MVLR) is modelled.	5		
b.	Explain how Logistic Regression is used for classification.	5		

UNIT - III

PART – A: (Short Answer Questions) (2 Marks)

		[CO#]	[PO#]
a.	What do you mean by Gradient Descent?		
b.	What are the conditions in which Gradient Descent is applied		
c.	Explain the importance of Stochastic Gradient Descent		
d.	Differentiate between Gradient Descent and Stochastic Gradient Descent		
e.	What is ensemble learning?		
f.	What is conditional Independence?		
g.	Why instance based learners are called lazy learners?		
h.	What is rule based ordering?		
i.	Distinguish between Coverage and Accuracy of a Rule.		
j.	List out the application areas of Case Based reasoning.		

PART – B: (Long Answer Questions) (10 Marks)

		Marks	[CO#]	[PO#]
1. a.	Explain how a rule based classifier works.	5		
b.	Give the Sequential Covering Algorithm for rule induction.	5		
2. a.	List out the advantages and disadvantages of KNN classifiers.	5		
b.	Explain with the help of a neat diagram about the working principle a Locally weighted Regression model.	5		
3. a.	What is Boosting? Discuss with neat relevant example?	5		
b.	Explain the concept of Bagging with its uses?	5		
4. a.	Explain boosting and ADA boosting algorithm with neat sketch?	5		
b.	What is Random forest? Explain with an example?	5		
5. a.	Explain the concept of Bayes theorem with an example.	5		
b.	Define (i) Prior Probability (ii) Conditional Probability (iii) Posterior Probability	5		

6. a.	Describe the concepts of Minimum Description Length principle.	5		
b.	Explain maximum likelihood algorithm	5		
7. a.	Illustrate the naïve Bayes classifier.	5		
b.	Explain naive Bayes classifier with example.	5		

UNIT - IV

PART – A: (Short Answer Questions) (2 Marks)

		[CO#]	[PO#]
a.	What is Artificial Neural Network?		
b.	What is linearly in-separable problem?		
c.	What are the types of problems in which Artificial Neural Network can be applied.		
d.	How do we define the stopping criteria of a Back Propagation algorithm.		
e.	Differentiate between local and global minima.		
f.	What is Squashing Function?		
g.	What is gradient descent and delta rule?		
h.	Why stochastic approximation to gradient descent is needed?		
i.	List some applications of ANNs.		
j.	What is the role bias in an ANN?		
k.	What is the goal of cluster analysis?		
l.	What is hierarchical clustering?		
m.	What are the different methods of hierarchical clustering?		

PART – B: (Long Answer Questions)

		Marks	[CO#]	[PO#]
1. a.	Explain the concept of a Perceptron with a neat diagram.	5		
b.	Explain how to learn Multilayer Networks using Backpropagation Algorithm.	5		
2. a.	What do you mean by Gradient Descent? What are the conditions in which Gradient Descent is applied.	5		
b.	What is radial basis function in neural network, explain in details?	5		
3. a.	Write the algorithm for Back propagation.	5		
b.	Derive the Backpropagation rule considering the training rule for Output Unit weights and Training Rule for Hidden Unit weights	5		
4. a.	What are the pros and cons of the K-Means algorithm? How do we select the number of clusters?	5		
b.	How can we make sure that K-Means output is not sensitive to initialization?	5		
5. a.	What is Density based clustering?	5		
b.	What are the advantages of the DBSCAN density-based Clustering Algorithm?	5		

6. a.	Describe Maximum Likelihood Hypothesis for predicting probabilities.	5		
b.	Explain the concept of EM Algorithm. Discuss what are Gaussian Mixtures	5		

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